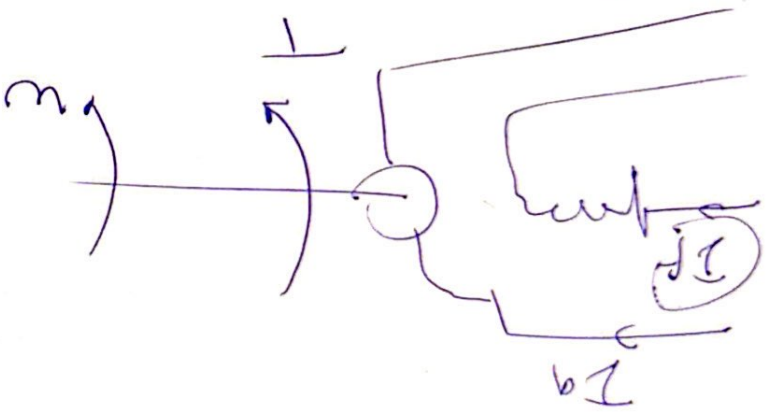
4F
D4

X	4F
Y	D4

X	DB	4F
Y	DB	D4

Code

```
MOV AL, 4F → X
ADD AL, D4 → Y
```



Scanned with CamScanner

Assembly

```
.Model small
.stack 64
.Data
```

Data Declaration

```
.code
Main Proc Far
Mov AX, @Data
Mov DS, AX
```

البرنامج

```
Mov AH, 4C
Int 21
EndP
Main
END Main
```

Scanned with CamScanner

Data Declaration

1-bit

4-bit ≡ nipple

define

DB Byte 8 bit ✓✓)10

DW word 16-bit ....

DD double word 32-bit ..... 4 location

DQ Quad. word 64 bit 8 location

DT 5 word ≡ 10 Byte 80 bit 10 location

KB 2^{10} B

MB 2^{20} B

GB 2^{30} B

TB 2^{40} B

line

24

2^{24}

$2^4 \cdot 2^{20}$

16 MB ✓

Scanned with CamScanner

(P) add 2 - Bytes Result ⟹ SUM

قيمة

X DB 4C

Y DB 3D

SUM DB ?

MOV AL, X

ADD AL, Y

MOV SUM, AL

X	4C
Y	3D
SUM	فاضي

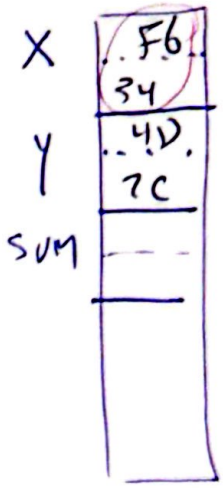
AL = 4C

AL = 4C + 3D

Scanned with CamScanner

P' Add 2 words

```
X    DW  34F6
Y    DW  2C4D
SUM  DW  ?
```



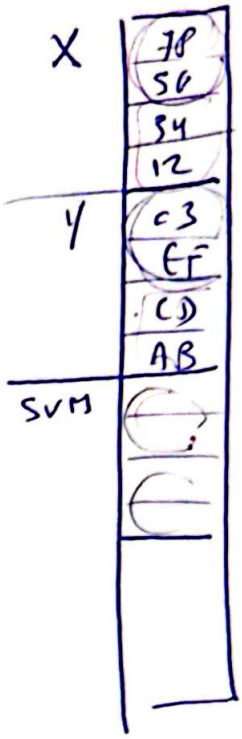

```
MOV  AX, X
ADD  AX, Y
MOV  SUM, AX
```

AX = 34F6

AX = 34F6 + 2C4D

P11 Add. 2 32 bit ~~Double word~~

X DD 1234 5678
Y DD ABCD EFC3
SUM DD ?




```
MOV AX, X
ADD AX, Y
MOV SUM, AX
```

```
MOV AX, X+2
ADC AX, Y+2
MOV SUM+2, AX
```

Scanned with CamScanner

JUMP

غير مشروط

JMP X1

X1

مشروط

* JC X1 if CF=1
* JNC CF=0

بعد ADD

* JZ ZF=1
* JNZ ZF=0

بعد DEC ++

---

* JE if AL=10
* JNE AL≠10
* JA AL > 10
* JB AL < 10
* JAE ≥
* JBE ≤

بعد CMP AL,10

لأرقام سالبة

JG >

JL <

Scanned with CamScanner

loop

لا بد Counter

عدد مرات التكرار

Mov CL , 4

Back

Dec CL

JNZ Back

Dec CL
JNZ Back = loop Back

Scanned with CamScanner

7th

P Add 6 Bytes ⟹ SUM

X DB 34, .., .., .., .., ..

SUM DW ?

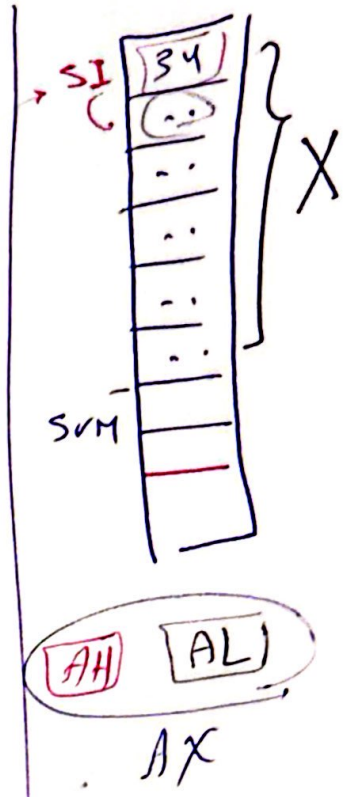
array LLB

```
MOV SI, offset X
```

بداية ال array

نظافة

```
MOV AL, 00
MOV AH, 00
MOV CL, 6
Back: ADD AL, [SI]
      JNC X1
      Inc AH
X1:   INC SI
      DEC CL
      JNZ Back
      MOV SUM, AX
```

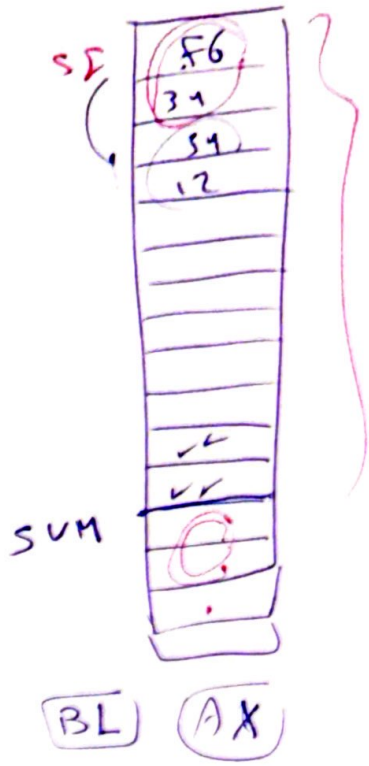


Scanned with CamScanner

P' add 6 words ⟶ sum

DataG DW 3456, 1234, ..., ..., ...

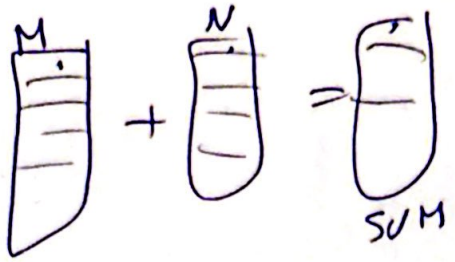
SUM DD ?




```
Mov AX,0000
Mov BL,00
Mov SI,offset DataG
Mov CL,6
Back: ADD AX,[SI]
      JNC skip
      Inc BL
skip: Inc SI
      Inc SI
      loop Back

      Mov SUM,AX
      Mov SUM+2,BL
```

P add 2 arrays (20 Byte)



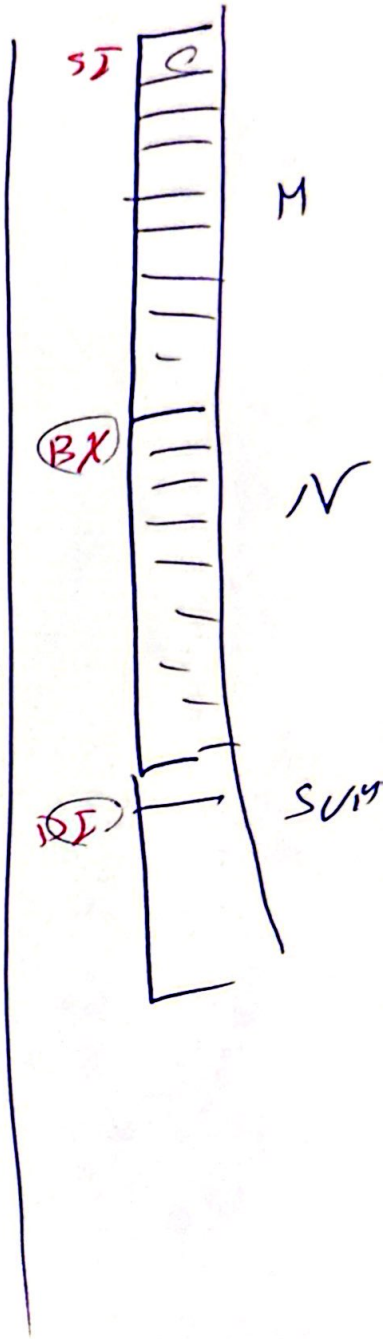
M DB ---
N DB ---
SUM DB 20 DUP(?)

MOV SI, offset M
- BX, - N
- DI, - SUM

~~MOV CL, 20~~

Ram
```
MOV AL, [SI]
ADD AL, [BX]
MOV [DI], AL
INC SI
INC BX
INC DI
```

Loop Back



Scanned with CamScanner